

OpenCPU SDK

Data API Notes

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This document is intended for system engineers (SEs), development engineers, and test engineers.

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Sales@neoway.com

Support@neoway.com

Website: <http://www.neoway.com>

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About This Document

Scope

This document is applicable to N706B and N520 modules.

Audience

This document is intended for [system engineers \(SEs\)](#), [development engineers](#), and [test engineers](#).

Change History

Issue	Date	Change	Changed By
1.0	2023-05	Initial release	Zhao Yating

Conventions

Symbol	Indication
	This warning symbol means danger. You are in a situation that could cause fatal device damage or even bodily damage.
	Means reader be careful. In this situation, you might perform an action that could result in module or product damages.
	Means note or tips for readers to use the module

1 Overview

This document introduces the APIs related to data operations in OpenCPU SDK, covering the following aspects:

- Profile Information
Used for querying and setting IP types, APNs, authentication types, usernames, and passwords.
- Auto-Redial Management
Manages the logic of automatic redialing.
- Dialing
Used for dialing to obtain an IP address and disconnecting the dialing connection.
- Network Sharing
Used for configuring network sharing types and DHCP-related parameters.

2 Data Structure

2.1 nwy_data_pdp_type_e

```
typedef enum {  
    NWY_DATA_PDP_TYPE_MIN = -1,           //IPv4 dial  
    NWY_DATA_PDP_TYPE_IPV4 = 1,           //IPv4 dial  
    NWY_DATA_PDP_TYPE_IPV6 = 2,           //IPv6 dial  
    NWY_DATA_PDP_TYPE_IPV4V6 = 3,         //IPv4/v6 dial  
    NWY_DATA_PDP_TYPE_PPP = 6,            //PPP dial  
} nwy_data_pdp_type_e;
```

2.2 nwy_data_auth_proto_e

```
typedef enum {  
    NWY_DATA_AUTH_PROTO_MIN = -1,  
    NWY_DATA_AUTH_PROTO_NONE = 0,  
    NWY_DATA_AUTH_PROTO_PAP,  
    NWY_DATA_AUTH_PROTO_CHAP,  
    NWY_DATA_AUTH_PROTO_PAP_CHAP  
} nwy_data_auth_proto_e;
```

2.3 nwy_data_call_state_e

```
typedef enum {  
    NWY_DATA_CALL_INVALID_STATE = -1,  
    NWY_DATA_CALL_DISCONNECTED_STATE = 0,  
    NWY_DATA_CALL_CONNECTED_STATE = 1  
} nwy_data_call_state_e;
```

2.4 nwy_data_type_e

```
typedef enum {  
    NWY_DATA_TYPE_NORMAL = 0x0,  
    NWY_DATA_TYPE_TETHER,           /* Network sharing, the host device obtains a LAN IP. */  
    NWY_DATA_TYPE_IPPT,             /* Network sharing, the host device obtains a WAN IP. */  
    NWY_DATA_TYPE_NORMAL_TETHER,    /* Network sharing, the host device obtains a LAN IP while  
                                     the internal protocol stack is enabled.*/  
    NWY_DATA_TYPE_NORMAL_IPPT,      /* Network sharing, the host device obtains a WAN IP while  
                                     the internal protocol stack is enabled.*/  
}
```

```
} nwy_data_type_e;
```

2.5 nwy_data_state_e

```
typedef enum {  
    NWY_DATA_STATE_INVALID = 0x0,  
    NWY_DATA_STATE_CONNECT,  
    NWY_DATA_STATE_DISCONNECT,  
    NWY_DATA_STATE_DISCONNECT_RETRY_MAX, /* The maximum number of auto-redial attempts has  
                                           been reached. */  
} nwy_data_state_e;
```

2.6 nwy_data_profile_info_t

```
typedef struct {  
    nwy_data_pdp_type_e pdp_type; /* PPP, IPv4,  
    char apn[NWY_APP_APN_MAX_LEN + 1]; /* Access Point Name  
    nwy_data_auth_proto_e auth_proto; /* authentication protocol  
    char user_name[NWY_APP_USER_MAX_LEN + 1]; /* user name  
    char pwd[NWY_APP_PWD_MAX_LEN + 1]; /* password  
} nwy_data_profile_info_t;
```

2.7 nwy_data_auto_connect_cfg_t

```
typedef struct {  
    uint16 retry_max; /* No further attempts will be made after reaching the maximum  
    reconnection attempts. */  
    uint8 interval[16]; /* The redial interval, in seconds, defaults subsequent elements to  
    the first non-zero value if user settings are incomplete. */  
} nwy_data_auto_connect_cfg_t;
```

2.8 nwy_data_start_call_t

```
typedef struct {  
    int cid;  
    nwy_data_call_action_type_e action;  
    nwy_data_trigger_type_e trigger_type;  
    nwy_data_call_auto_type_e call_auto_type;  
    nwy_data_profile_info_t set_profile;  
    int if_internal_call;  
} nwy_data_start_call_t;
```

2.9 nwy_in4_addr_t

```
typedef struct {  
    uint32 s_addr;  
}nwy_in4_addr_t;
```

2.10 nwy_in6_addr_t

```
typedef struct {  
    uint8 u6_addr8[16];  
    uint16 u6_addr16[8];  
    uint32 u6_addr32[4];  
}nwy_in6_addr_t;
```

2.11 v4_info_t

```
typedef struct {  
    nwy_in4_addr_t public_ip;           /**<IP address. */  
    nwy_in4_addr_t primary_dns;         /**<Primary domain name service (DNS) IP address. */  
    nwy_in4_addr_t secondary_dns;       /**<Secondary DNS IP address. */  
}v4_info_t;
```

2.12 v6_info_t

```
typedef struct {  
    nwy_in6_addr_t public_ip_v6;        /**< IPv6 address. */  
    nwy_in6_addr_t primary_dns_v6;      /**<Primary domain name service (DNS) IPv6 address. */  
    nwy_in6_addr_t secondary_dns_v6;    /**<Secondary DNS IPv6 address. @newpagetable */  
}v6_info_t;
```

2.13 nwy_data_callinfo_t

```
typedef struct {  
    char ipv4_str[NWY_APP_IP_STR_MAX_LEN];  
    char ipv6_str[NWY_APP_IP_STR_MAX_LEN];  
    nwy_data_call_state_e state;  
    nwy_data_profile_info_t profile;  
}nwy_data_callinfo_t;
```

2.14 nwy_data_dhcp_t

```
typedef struct {  
    char gw[20];  
    char netmask[20];  
    char start_ip[20];  
    char end_ip[20];  
    unsigned int lease_time;  
}nwy_data_dhcp_t;
```

2.15 nwy_data_cb_func

```
typedef void (*nwy_data_cb_func)  
(  
    int profile_idx,  
    nwy_data_call_state_e ind_state  
);
```

3 APIs

This section introduces the device management APIs.

The API definitions can be found in **nwy_common.h** and **nwy_dm.h**.

3.1 nwy_data_profile_set

Function	<code>nwy_error_e nwy_data_profile_set(nwy_sim_id_e sim_id, int profile_idx, nwy_data_profile_info_t *profile_info);</code>
Description	Sets information such as IP type, APN, and authentication.
Parameter	<code>sim_id</code> : SIM card information. <code>profile_idx</code> : CID (Connection Identifier). <code>profile_info</code> : Structure containing the parameters to be set.
Return Value	Successful: <code>NWY_SUCCESS</code> Failed: another value

3.2 nwy_data_profile_get

Function	<code>nwy_error_e nwy_data_profile_get(nwy_sim_id_e sim_id, int profile_idx, nwy_data_profile_info_t *profile_info);</code>
Description	Retrieves information such as IP type, APN, and authentication.
Parameter	<code>sim_id</code> : SIM card information. <code>profile_idx</code> : CID (Connection Identifier). <code>profile_info</code> : Structure to store the retrieved information.
Return Value	Successful: <code>NWY_SUCCESS</code> Failed: another value

3.3 nwy_data_reg_cb

Function	<code>nwy_error_e nwy_data_reg_cb(nwy_sim_id_e sim_id, int profile_idx, nwy_data_cb_func cb_func);</code>
Description	Registers a callback function for dialing operations.

Parameter	sim_id: SIM card information.
	profile_idx: CID (Connection Identifier).
	cb_func: Callback function to be registered.
Return Value	Successful: NWY_SUCCESS
	Failed: another value

3.4 nwy_data_unreg_cb

Function	nwy_error_e nwy_data_unreg_cb(nwy_sim_id_e sim_id, int profile_idx, nwy_data_cb_func cb_func);
Description	Unregisters a callback function for dialing operations.
Parameter	sim_id: SIM card information.
	profile_idx: CID (Connection Identifier).
	cb_func: Callback function to be unregistered.
Return Value	Successful: NWY_SUCCESS
	Failed: another value

3.5 nwy_data_auto_connect_set

Function	nwy_error_e nwy_data_auto_connect_set(int cid, nwy_data_auto_connect_cfg_t *cfg);
Description	Sets control parameters for the auto-redial feature.
Parameter	cfg: Structure containing control parameters for the auto-redial feature.
Return Value	Successful: NWY_SUCCESS
	Failed: another value

3.6 nwy_data_call_start

Function	nwy_error_e nwy_data_call_start(nwy_sim_id_e sim_id, nwy_data_start_call_t *param);
Description	Activates the specified dialing connection.
Parameter	sim_id: SIM card information.
	param: Parameters required to activate the dialing connection.
Return Value	Successful: NWY_SUCCESS
	Failed: another value

3.7 nwy_data_call_stop

Function	<code>nwy_error_e nwy_data_call_stop(nwy_sim_id_e sim_id, int profile_idx);</code>
Description	Deactivates the specified dialing connection.
Parameter	<code>sim_id</code> : SIM card information. <code>profile_idx</code> : CID (Connection Identifier).
Return Value	Successful: <code>NWY_SUCCESS</code> Failed: another value

3.8 nwy_data_call_info_get

Function	<code>nwy_error_e nwy_data_call_info_get(int profile_idx, nwy_data_callinfo_t * info_ptr);</code>
Description	Queries the connection information for a specific CID.
Parameter	<code>profile_idx</code> : CID (Connection Identifier). <code>info_ptr</code> : Structure to store the retrieved connection information.
Return Value	Successful: <code>NWY_SUCCESS</code> Failed: another value

3.9 nwy_data_dhcp_set

Function	<code>nwy_error_e nwy_data_dhcp_set(nwy_data_dhcp_t *dhcp);</code>
Description	Sets DHCP parameters for network sharing.
Parameter	<code>dhcp</code> : Structure containing the DHCP parameters to be set.
Return Value	Successful: <code>NWY_SUCCESS</code> Failed: another value

3.10 nwy_data_dhcp_get

Function	<code>nwy_error_e nwy_data_dhcp_get(nwy_data_dhcp_t *dhcp);</code>
Description	Retrieves DHCP parameters for network sharing.
Parameter	<code>dhcp</code> : Structure to store the retrieved DHCP parameters.
Return Value	Successful: <code>NWY_SUCCESS</code> Failed: another value

3.11 nwy_data_type_set

Function	nwy_error_e nwy_data_type_set(int profile_idx, nwy_data_type_e type);
Description	Sets the type of dialing connection.
Parameter	profile_idx: CID (Connection Identifier). type: Dialing connection type.
Return Value	Successful: NWY_SUCCESS Failed: another value